

PROJECT AND TEAM INFORMATION

Project Title

Disaster Management System

Student / Team Information

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PROPOSAL DESCRIPTION

Motivation

Natural disasters like floods, earthquakes, landslides, and fires can impact many individuals and complicate the organization of information regarding the affected regions, individuals, rescue teams, and vital resources. Managing large volumes of data manually becomes messy and inefficient very quickly. We built this Disaster Management System to give administrators a clear, organized way to store and update critical disaster response records. The system tracks event data, victim details, rescue team assignments, emergency aid requests, and location mapping.

Beyond its practical application, this project applies concepts learned in our 3rd semester Data Structures and C++ course to a real-world scenario. We implemented foundational concepts—including arrays, linked lists, stacks, queues, graphs, search and sort algorithms, and C++ file handling—to structure and process incoming data efficiently.

State of the Art / Current solution

Currently, disaster management requires cooperation of different government agencies, emergency response teams, communication systems, databases, and geographical information system (GIS). Modern disaster management relies heavily on interconnected government networks, GIS mapping, complex databases, and emergency response infrastructure. While these enterprise tools handle massive workloads, they often carry high complexity and depend on specialized software.

Because full-scale corporate platforms are too vast for a baseline student project, our system focuses on the core mechanics. Built in C++, it offers a streamlined approach to managing victim databases, dispatching resources, and organizing rescue routes using essential Data Structures and Algorithms (DSA).

Project Goals and Milestones

Main Goal:

The primary aim of this project is to create an easy-to-use Disaster Management System utilizing C++ and DSA. This system should effectively handle records of disasters, track affected individuals, coordinate rescue teams, manage resources, process emergency requests, and pinpoint locations.

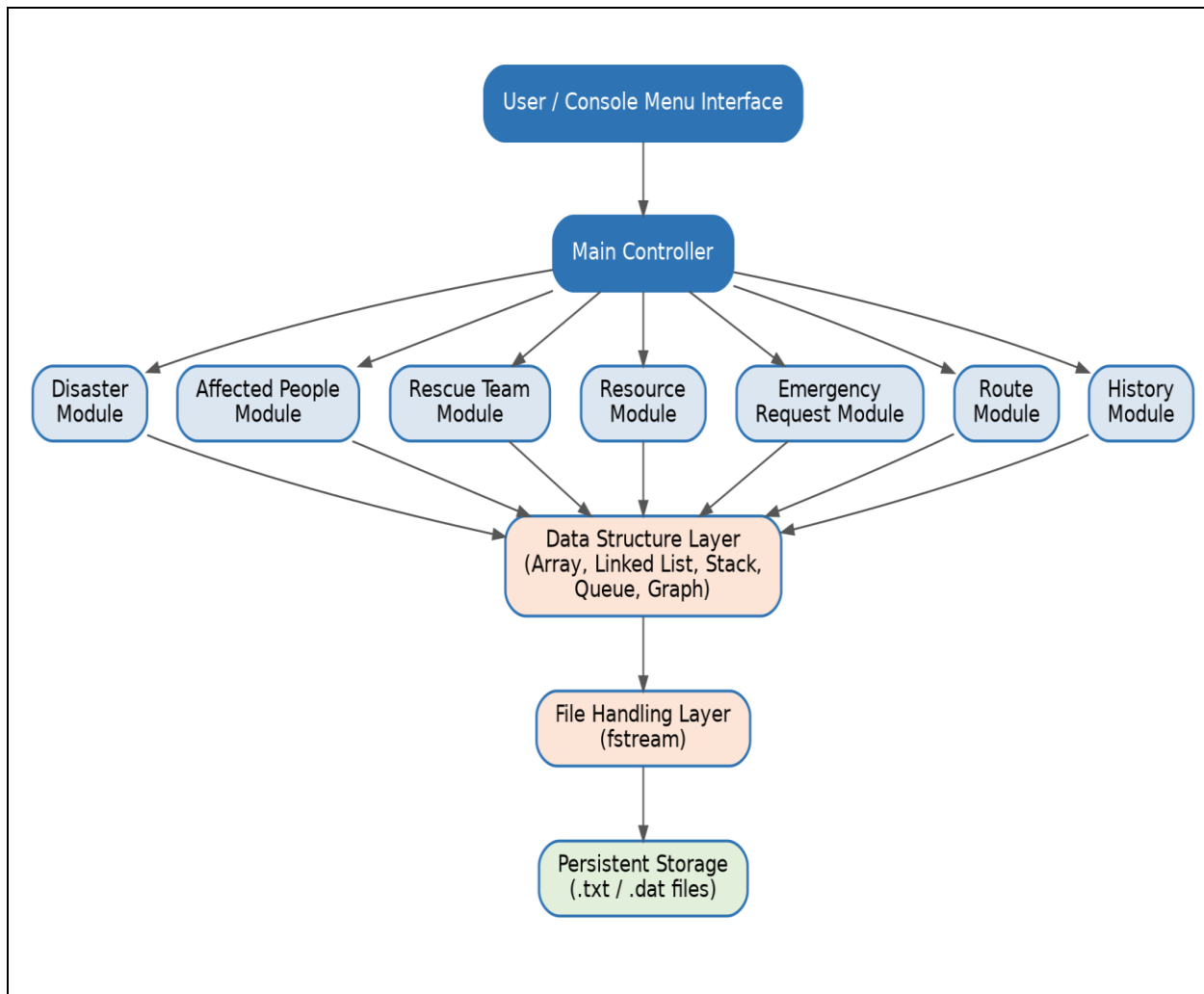
Milestones:

- Identify the requirements and major modules of the system.
- Design the data structures required for storing disaster-related information.
- Implement arrays and linked lists for managing records.
- Implement stack and queue for managing operations and emergency requests.
- Implement graphs with BFS/DFS for representing and exploring locations.
- Implement searching and sorting for efficient record management.
- Implement file handling in C++ to save and retrieve records.
- Integrate all modules into a single working system.
- Test the system with different disaster scenarios and inputs.
- Prepare the final documentation and project demonstration.

Project Approach

The application will be created as a console-based program utilizing C++ along with basic Data Structures and Algorithms. It will consist of several modules designed to handle disaster-related data, including information about affected individuals, rescue teams, available resources, emergency requests, and various locations. To store and manage records effectively, arrays and linked lists will be employed, while stacks and queues will organize recent actions and emergency requests. Graphs will illustrate locations and the roads connecting different regions, and BFS and DFS algorithms will be applied to explore these connected sites. Efficiently locating and ordering records will be supported by searching and sorting algorithms. To ensure that vital information is permanently stored and easily retrieved, file handling techniques will be incorporated. Key C++ concepts such as classes, objects, constructors, functions, inheritance, and encapsulation will guide the development of the distinct modules. Following their individual creation, these modules will be combined into a unified system and subjected to tests involving various disaster scenarios to verify functionality and showcase the practical use of DSA principles.

System Architecture (High Level Diagram)



Project Outcome / Deliverables

The outcome of the project will be a fully functional Disaster Management System developed in C++. This system will showcase how Data Structures and Algorithms can be applied to solve real-world issues. It will serve as a well-organized platform to manage records of disasters, information about affected individuals, rescue teams, resources, emergency requests, and geographical locations. The system will utilize various data structures such as arrays, linked lists, stacks, queues, graphs, as well as algorithms for BFS/DFS, searching, and sorting to efficiently store, process, and retrieve data. Essential file handling techniques will be implemented to ensure important records can be saved and accessed later. Key deliverables will consist of the operational C++ application, complete source code, system architecture, module designs, test cases along with results, project documentation, and a final demonstration of the project. Additionally, the project will illustrate how fundamental DSA concepts can be integrated with C++ programming to create a well-organized and functional disaster management solution.

Assumptions

The proposed Disaster Management System assumes that the system will be operated by authorized users, such as disaster-management staff, volunteers, or administrators, who will enter and update the required information. It is assumed that the data provided by users, including disaster details, affected people, rescue teams, resources, and locations, is accurate and relevant. The system will operate as a C++ console-based application on a computer without requiring an internet connection or external services. Location and road information will be entered manually and will be considered valid for the purpose of route management. The project will focus on demonstrating the practical use of B.Tech 3rd-semester DSA concepts, including arrays, linked lists, stacks, queues, graphs, BFS/DFS, searching, and sorting. Real-time GPS tracking, live weather updates, satellite data, and external APIs are outside the scope of the initial project.

References

- GeeksforGeeks — Data Structures (Arrays, Linked List, Stack, Queue, Graph, BFS/DFS): <https://www.geeksforgeeks.org/data-structures/>
- cppreference.com — C++ Standard Library documentation (stack, queue, fstream): <https://en.cppreference.com/>
- Cormen, Leiserson, Rivest, Stein — Introduction to Algorithms (graph traversal, searching, sorting).
- National Disaster Management Authority (NDMA) resources on disaster response coordination: <https://ndma.gov.in/>